

Foundation (Jalapeno)

- Q1.** If the slope of a line is 3, what is the slope of a line perpendicular to it?
(A) -3
(B) $\frac{1}{3}$
(C) $-\frac{1}{3}$
(D) 3
(E) 0
- Q2.** What is the equation of a line perpendicular to $y = 2x + 1$ and passing through $(0, 0)$?
(A) $y = -\frac{1}{2}x$
(B) $y = \frac{1}{2}x$
(C) $y = 2x$
(D) $y = -2x$
(E) $y = x$
- Q3.** In the stock market, the price of a stock is modeled by $y = 4x + 5$. What is the slope of a line perpendicular to this model?
(A) -4
(B) $\frac{1}{4}$
(C) $-\frac{1}{4}$
(D) 4
(E) 0
- Q4.** Two lines have slopes m_1 and m_2 . If $m_1 = 2$ and the lines are perpendicular, what is m_2 ?
(A) -2
(B) $\frac{1}{2}$
(C) $-\frac{1}{2}$
(D) 2
(E) 0
- Q5.** A savings account earns interest according to the equation $y = 5x + 1000$, where y is the amount of money after x years. What is the slope of a line perpendicular to this equation?
(A) -5
(B) $\frac{1}{5}$
(C) $-\frac{1}{5}$
(D) 5
(E) 0
- Q6.** The tax on a certain investment is represented by $y = 3x - 2$. What is the equation of a line perpendicular to this tax model and passing through $(1, 1)$?
(A) $y = -\frac{1}{3}x + \frac{4}{3}$
(B) $y = \frac{1}{3}x + \frac{2}{3}$
(C) $y = -3x + 2$
(D) $y = 3x - 2$
(E) $y = x - 1$
- Q7.** The equation $y = -2x + 3$ represents the price of a stock in terms of time x . What is the equation of a line perpendicular to this equation and passing through the point $(2, 0)$?
(A) $y = \frac{1}{2}x - 1$
(B) $y = -\frac{1}{2}x + 2$
(C) $y = 2x - 3$
(D) $y = -2x + 3$
(E) $y = x - 2$
- Q8.** If two lines have slopes m_1 and m_2 and are perpendicular, which of the following is true?
(A) $m_1 + m_2 = 0$
(B) $m_1 \cdot m_2 = 0$
(C) $m_1 \cdot m_2 = 1$
(D) $m_1 \cdot m_2 = -1$
(E) $m_1 = m_2$
- Q9.** A savings account earns interest according to the equation $y = 5x + 1000$, where y is the amount of money after x

Open Ended Questions (FRQ)

- Q9.** Show that the lines $y = 2x + 1$ and $y = -\frac{1}{2}x - 3$ are perpendicular. Then, find the equation of a line that is perpendicular to both and passes through the point of intersection of the two given lines.
- Q10.** A financial analyst uses the equation $y = 3x + 2$ to model the growth of an investment. Find the equation of a line perpendicular to this model and passing through the point $(1, 5)$. Then, explain what it means for two lines to be perpendicular in the context of this financial model.

Chef's Tips

- Chef's Tips
- For multiple-choice questions, ensure students understand the concept of negative reciprocal slopes.
- For free-response questions, emphasize the importance of showing work and explaining the concept of perpendicularity in the given context.